Slow Momentum with Fast Reversion

IEOR 4576 - Executed replication

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Wood, Roberts & Zohren (2022), arXiv:2105.13727v3

Research question. Does adding online Gaussian-process changepoint location and severity to a Deep Momentum Network improve out-of-sample risk-adjusted returns, and does any gain survive trading costs?

Scope. This is an executed WRDS adaptation of the original 50-futures study, with 50 search candidates per model/window. It uses real historical settlements and fitted models. Pinnacle histories and exact session/roll conventions are not available, so numerical identity is not claimed.

The mean five-window gross Sharpe is 1.815 for LSTM and 1.602 for the validation-selected CPD model (difference -0.212). The published figures are 1.62 and 2.16. The datasets, roll conventions and numerical backends differ, so their numerical difference is not an implementation error estimate.

At 2.5 one-way bps, original-style turnover-adjusted mean Sharpe is LSTM: 0.302, LSTM-CPD-Optimised: -0.377, TSMOM w=0: 0.940. This is a cost scenario, not an estimate of broker fees or market impact.

Using un-winsorized realized returns and charging both roll legs at 2.5 bps, mean net Sharpe is LSTM: 0.257, LSTM-CPD-Optimised: -0.418, TSMOM w=0: 0.860. These figures assess tradability separately from the author-style return labels; daily quotes still do not establish executable spreads or capacity.

Evidence. 50 mapped instruments, 4,321,070 raw contract-day observations and 346,218 feature rows. Completed model/window experiments: 36; random-search fits: 1800.

The professor's feedback on reversal costs and futures rolls is addressed after the original-style gross experiments. All reported tables and figures are generated by reusable Python modules, with an executed notebook and one-command rebuild.

Methodology and chronology

Returns and rolls. Five-business-day calendar roll before vendor notice/last-trade dates; select the nearest maturity using prices known at the decision date. Compute P&L within the held contract, never across a near/far price difference. Missing composite calendars for three markets use the vendor's same-month pit-contract schedule. Causal stale settlement marks are counted in the coverage table; these are valuations rather than executable fills.

Features. Execute the preserved author feature functions with pandas API compatibility changes: normalized 1/21/63/126/252-session returns and three MACD pairs, price-index EWM winsorization at five standard deviations, and 60-session EWM volatility. The original-style labels are returns of the winsorized index; raw-return tradability sensitivity is separate.

GP-CPD. Independently fit stationary and changepoint Matérn 3/2 GP kernels on every trailing 10/21/63/126/252-session window. Use maximum 200 L-BFGS-B iterations and stationary-parameter initialization of the CP model. Exact state-space likelihood evaluation matches the dense covariance. Severity is sigmoid(NLML_M - NLML_CP); relative location follows the author code.

LSTM. Stateless 63-step LSTM, tanh positions, input/output dropout, pooled negative Sharpe loss, Adam, parameter-wise gradient clipping and output max-norm. Non-overlapping training/validation sequences, chronological per-instrument 90/10 split, purged boundary labels, maximum 300 epochs and patience 25. The CPU/Metal PyTorch port preserves these components, with backend numerical differences disclosed.

Model selection. 50 distinct deterministic random draws from the published hyperparameter grid per model/window. CPD horizon selection uses only pre-test validation loss. One complete search repetition, base seed 67; repeated-seed robustness is not claimed. Compatible earlier CPU candidates retain their original provenance.

Backtest. Expand training from 1990. Main OOS windows: 1995-1999, 2000-2004, 2005-2009, 2010-2014, 2015-2019. A separate pre-2020 refit scores 2020. Exclude the one-year 2020 window from the five-window exhibit means. Eligibility uses prior validation availability and Appendix A's first-test dates.

Portfolio. Equal 1/N contribution across eligible instruments, with per-asset 15% annualized ex-ante risk scaling. Calendar differences contribute cash on unavailable-market dates. No hindsight volatility scaling is used for raw output; Exhibit 4 uses a single author-style factor targeting 15% mean window volatility; this is ex-post presentation only.

Exhibits 3 and 4

Raw signal output

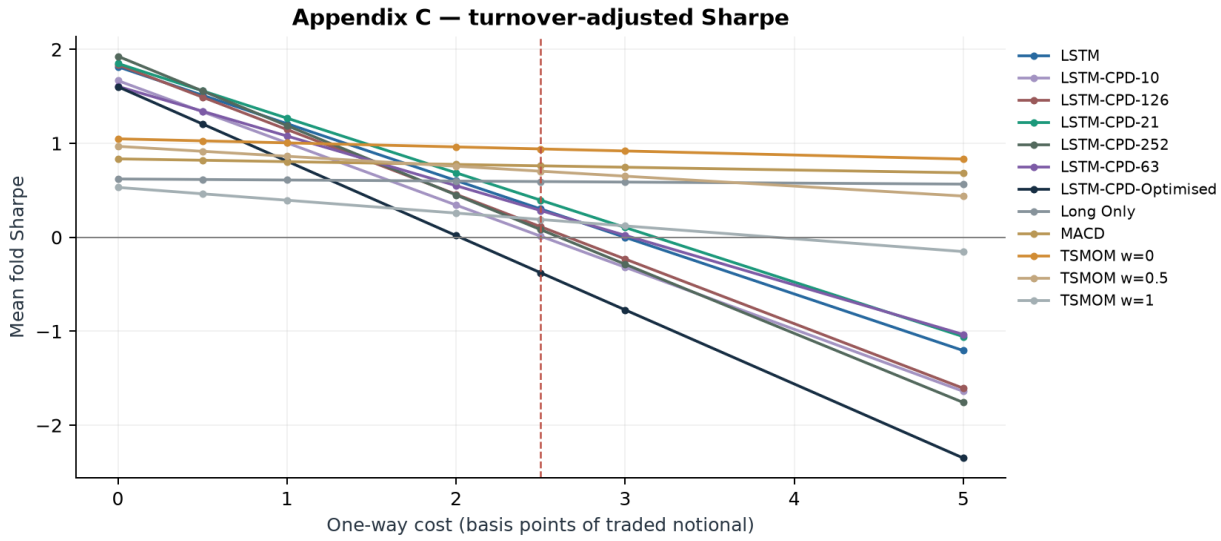
Strategy	CAGR	Vol.	Sharpe	MDD
LSTM	2.71%	1.55%	1.815	1.89%
LSTM-CPD-10	2.86%	2.00%	1.667	2.35%
LSTM-CPD-126	3.46%	2.11%	1.836	3.13%
LSTM-CPD-21	3.41%	2.24%	1.850	2.80%
LSTM-CPD-252	3.71%	1.91%	1.927	2.14%
LSTM-CPD-63	2.91%	2.24%	1.606	3.24%
LSTM-CPD-Optimised	3.02%	1.88%	1.602	2.49%
Long Only	2.95%	4.91%	0.621	11.76%
MACD	2.60%	3.29%	0.835	4.97%
TSMOM w=0	4.37%	4.35%	1.048	6.27%
TSMOM w=0.5	3.27%	3.47%	0.968	5.15%
TSMOM w=1	2.12%	4.33%	0.531	7.23%

Ex-post 15% mean-window volatility rescaling

Strategy	CAGR	Vol.	Sharpe	MDD
LSTM	29.08%	15.00%	1.815	16.99%
LSTM-CPD-10	23.05%	15.00%	1.667	16.24%
LSTM-CPD-126	26.83%	15.00%	1.836	20.12%
LSTM-CPD-21	24.46%	15.00%	1.850	17.30%
LSTM-CPD-252	32.84%	15.00%	1.927	15.87%
LSTM-CPD-63	20.78%	15.00%	1.606	20.52%
LSTM-CPD-Optimised	25.96%	15.00%	1.602	18.43%
Long Only	8.50%	15.00%	0.621	32.07%
MACD	11.70%	15.00%	0.835	21.38%
TSMOM w=0	15.21%	15.00%	1.048	20.30%
TSMOM w=0.5	14.21%	15.00%	0.968	20.63%
TSMOM w=1	6.82%	15.00%	0.531	23.41%

All entries average the corresponding metric across the five main OOS windows. Full downside deviation, Sortino, Calmar, hit rate and profit/loss ratios are in the executed notebook and CSV tables. The separate 2020 test is reported in the regime diagnostics.

Trading costs and futures rolls

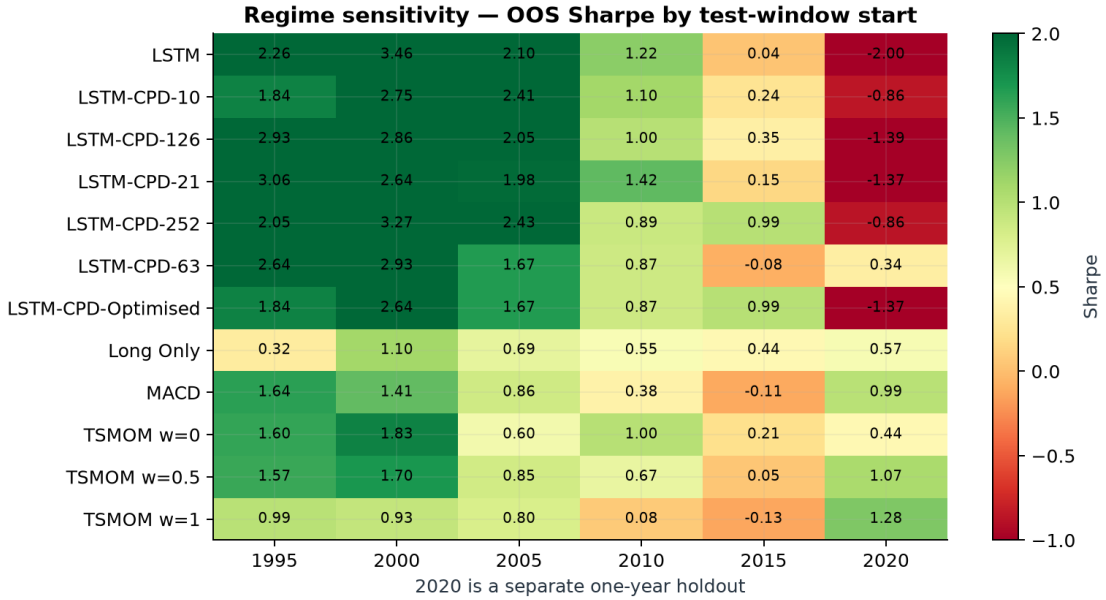


Appendix C charges one-way traded-notional costs, including initial entry, exit, reversals and volatility-driven resizing. The 2.5-bps reference addresses the graded-proposal feedback. It is a scenario; daily settlements do not identify spreads, impact or capacity.

Raw-return + roll scenario	Net Sharpe	Turnover/year	Roll extra/year
LSTM	0.257	96.9	2.8
LSTM-CPD-21	0.348	115.2	4.9
LSTM-CPD-63	0.236	109.2	4.6
LSTM-CPD-Optimised	-0.418	149.3	3.0
TSMOM w=0	0.860	32.1	13.7

Both legs of a roll incur fees. A futures roll does not exchange the full principal of either contract: marking the same contract over each return interval prevents the near/far basis from becoming a fictitious cash flow. Positions are expressed in fractional risk-scaled notional, not integer contracts. A one-session signal delay and break-even costs are also reported.

Risk, interpretation and reproducibility



Regime sensitivity and drawdowns are reported alongside the aggregate Sharpe ratios. The 2020 column is a separate one-year test. A paired 63-session moving-block bootstrap measures conditional return-series uncertainty, not training-seed or model-selection uncertainty.

Limitations. WRDS replaces Pinnacle; several markets start later than the paper, NR ends in 2015, and the euro has no pre-euro extension. Local-currency return normalization, stale marks, idealized close execution and one search repetition limit numerical comparability. No separate MACD ablation or measured market-impact calibration is claimed.

Rebuild. From the prepared project environment, run `make reproduce`. The main configuration uses the published 50-candidate search budget. Use `make fresh` to train all candidates from random initialization.

Notebook: report/Paper67_Original_Study_Replication.ipynb
Tables and figures: results/replication/ and results/extension/
Provenance: report/execution_manifest.json, UPSTREAM.json and raw WRDS manifests

Sources: Wood, Roberts & Zohren (2022), arXiv:2105.13727v3; kieranjwood/slow-momentum-fast-reversion; LSTM/CPD components of kieranjwood/trading-momentum-transformer. Course guidelines and graded proposal were supplied locally. Author MIT licenses are preserved.